

SEVERITY & CASE CLASSIFICATION

Adverse Drug Reactions in Pharmacovigilance

Assignment: Classification of adverse drug reactions based on seriousness, severity and expectedness, with evaluation of clinical impact using pharmacovigilance principles.

Academic Assignment	Pharmacovigilance / Drug Safety
Topic	Severity & Case Classification
Format	Internship Task Submission

Note: This assignment intentionally contains no personal information.

1. Introduction

Pharmacovigilance is the science and activities concerned with the detection, assessment, understanding and prevention of adverse effects or other medicine-related problems. Proper case classification helps drug-safety professionals determine the clinical importance of a reported event, prioritize follow-up, identify safety signals, and communicate risks appropriately.

For an individual adverse event, several concepts should be kept separate. **Severity** describes the intensity of the event, while **seriousness** is a regulatory classification based on outcome or important medical circumstances. **Expectedness** asks whether the event is consistent with the applicable product information or reference safety information. These dimensions should not be treated as interchangeable.

2. Objectives

- Define severity, seriousness and expectedness in the context of adverse drug reactions (ADRs).
- Differentiate serious from non-serious adverse events using standard pharmacovigilance criteria.
- Classify hypothetical cases according to clinical impact and expectedness.
- Demonstrate a structured approach to case assessment and documentation.
- Recognize when an event may require expedited reporting or additional medical review.

3. Key Pharmacovigilance Definitions

Term	Meaning
Adverse Event (AE)	Any untoward medical occurrence after use of a medicinal product; a causal relationship does not have to be established.
Adverse Drug Reaction (ADR)	A harmful and unintended response to a medicinal product where a causal relationship is at least a reasonable possibility.
Severity	The intensity of an event, commonly described as mild, moderate or severe.
Seriousness	A regulatory concept based on outcomes or important circumstances, such as death, life-threatening condition, hospitalization, disability, congenital anomaly, or another medically important condition.
Expectedness	Whether the nature or specificity of the event is consistent with the applicable reference safety information/product labeling.
Clinical Impact	The effect of the event on the patient's health, treatment, monitoring, functional status and need for medical intervention.

4. Severity Classification

Category	General Description	Illustrative Example
Mild	Transient or easily tolerated symptoms with little or no limitation of normal activity.	Mild headache after a dose, resolving without treatment.
Moderate	Noticeable symptoms that may interfere with normal activity and may require symptomatic treatment.	Persistent nausea requiring an antiemetic.
Severe	Marked intensity or substantial functional impairment; may require urgent medical management.	Severe bronchospasm requiring emergency treatment.

Important: Severe does not automatically mean serious. For example, a severe headache may not meet a seriousness criterion, whereas a mild event can be serious if it results in hospitalization or another qualifying outcome.

5. Seriousness Classification

Serious Criterion	Assessment Principle
Death	The event resulted in death.
Life-threatening	The patient was at immediate risk of death at the time of the event; it does not mean that the event could theoretically have become fatal.
Hospitalization	The event caused inpatient hospitalization or prolonged an existing hospitalization.
Persistent/significant disability or incapacity	The event resulted in substantial disruption of a person's ability to conduct normal life functions.
Congenital anomaly/birth defect	The event resulted in a congenital anomaly or birth defect.
Other medically important event	An important medical event that may jeopardize the patient or require intervention to prevent one of the outcomes above.

6. Expectedness Classification

Expectedness is assessed against the current applicable reference safety information, such as the approved product information or investigator reference safety information, depending on the reporting context. An event is generally considered **expected** when its nature and specificity are consistent with that reference information. An event may be considered **unexpected** when its nature, severity or specificity is not consistent with the applicable reference information.

Expectedness should therefore not be decided solely from whether the reaction is common or uncommon. The exact reference document, version and relevant wording should be checked during an actual pharmacovigilance case assessment.

7. Case-Based Evaluation

The following hypothetical cases demonstrate how severity, seriousness, expectedness and clinical impact can be assessed. They are educational examples rather than real patient cases.

Case 1 – Mild headache

Scenario: A patient develops a mild headache two hours after taking a prescribed medicine. The headache resolves spontaneously the same day and no treatment or hospitalization is required.

Classification: Severity: Mild | Seriousness: Non-serious | Expectedness: Expected, if listed in the applicable reference information.

Clinical impact: Low clinical impact; routine documentation and appropriate monitoring are generally sufficient.

Case 2 – Persistent vomiting

Scenario: A patient experiences repeated vomiting after treatment and requires an antiemetic and additional clinical observation, but does not require admission to hospital.

Classification: Severity: Moderate | Seriousness: Non-serious, assuming no qualifying serious criterion is met | Expectedness: Determined by the applicable reference information.

Clinical impact: Moderate clinical impact because treatment is required and adherence/hydration may be affected.

Case 3 – Anaphylaxis requiring emergency care

Scenario: Shortly after administration, a patient develops anaphylaxis requiring emergency treatment with appropriate supportive measures and close monitoring.

Classification: Severity: Severe | Seriousness: Serious (life-threatening and/or medically important, depending on documented circumstances) | Expectedness: Check the applicable reference information.

Clinical impact: High clinical impact; urgent medical intervention is required and the case warrants careful safety assessment.

Case 4 – Hospitalization for severe skin reaction

Scenario: A patient develops an extensive drug-related skin reaction and is admitted to hospital for treatment.

Classification: Severity: Severe | Seriousness: Serious because hospitalization occurred | Expectedness: Check the applicable reference information.

Clinical impact: High clinical impact due to inpatient care and potential risk of complications.

Case 5 – Transient injection-site pain

Scenario: A patient reports localized pain at the injection site. Symptoms are mild and disappear within 24 hours.

Classification: Severity: Mild | Seriousness: Non-serious | Expectedness: Expected if consistent with the applicable reference information.

Clinical impact: Low clinical impact; no major intervention or change in treatment is required.

Case 6 – Medically important arrhythmia

Scenario: A patient develops a clinically significant arrhythmia after starting therapy. The patient is treated urgently to prevent deterioration but is not admitted to hospital.

Classification: Severity: Severe | Seriousness: Potentially serious as an 'other medically important' event, based on medical judgment and documented circumstances | Expectedness: Check the applicable reference information.

Clinical impact: Potentially high clinical impact because urgent intervention is needed to prevent a serious outcome.

8. Case Classification Summary

Case	Severity	Seriousness	Expectedness	Clinical impact
1	Mild	Non-serious	Expected*	Low
2	Moderate	Non-serious*	Reference check	Moderate
3	Severe	Serious	Reference check	High
4	Severe	Serious	Reference check	High
5	Mild	Non-serious	Expected*	Low
6	Severe	Potentially serious	Reference check	Potentially high

*Expectedness is illustrative only. In an actual case, the current applicable reference safety information must be reviewed.

9. Practical Case Assessment Workflow

1. Identify the event and obtain the relevant clinical description.
2. Assess the intensity/severity of the event: mild, moderate or severe.
3. Determine whether any regulatory seriousness criterion is met.
4. Check the applicable reference safety information to assess expectedness.
5. Assess the clinical impact, treatment required, outcome and relevant risk factors.
6. Document the rationale for each classification clearly and consistently.
7. Determine whether follow-up information or expedited reporting requirements apply under the relevant local/regional rules.

10. Important Distinctions in Pharmacovigilance

- **Severity ≠ Seriousness:** intensity and regulatory seriousness are different concepts.
- **Expected ≠ Non-serious:** an expected reaction can be serious, and an unexpected event can be non-serious.
- **Clinical judgment matters:** medically important events may qualify as serious even without hospitalization.
- **Reference information is essential:** expectedness must be assessed against the appropriate current reference document.
- **Documentation should be traceable:** the case narrative and assessment should support the classification assigned.

11. Conclusion

Accurate classification of adverse drug reactions is a fundamental pharmacovigilance activity. Severity describes the intensity of an event, seriousness identifies regulatory significance, and expectedness compares the event with the applicable reference safety information. A structured assessment of these dimensions helps identify clinically important cases, supports consistent safety reporting, and contributes to the ongoing evaluation of medicine safety.

12. Suggested References

- U.S. Food and Drug Administration (FDA). Guidance for Industry: E2A Clinical Safety Data Management: Definitions and Standards for Expedited Reporting.
- International Council for Harmonisation (ICH). E2A: Clinical Safety Data Management — Definitions and Standards for Expedited Reporting.
- International Council for Harmonisation (ICH). E2D: Post-Approval Safety Data Management: Definitions and Standards for Management and Reporting of Individual Case Safety Reports.
- World Health Organization (WHO). Pharmacovigilance: ensuring the safe use of medicines.
- European Medicines Agency (EMA). Guideline on good pharmacovigilance practices (GVP), Module VI: Management and reporting of adverse reactions to medicinal products.

Academic note: This document is prepared as an educational internship assignment. Actual regulatory reporting decisions should follow the requirements of the applicable authority, reporting system, product information and pharmacovigilance standard operating procedures.